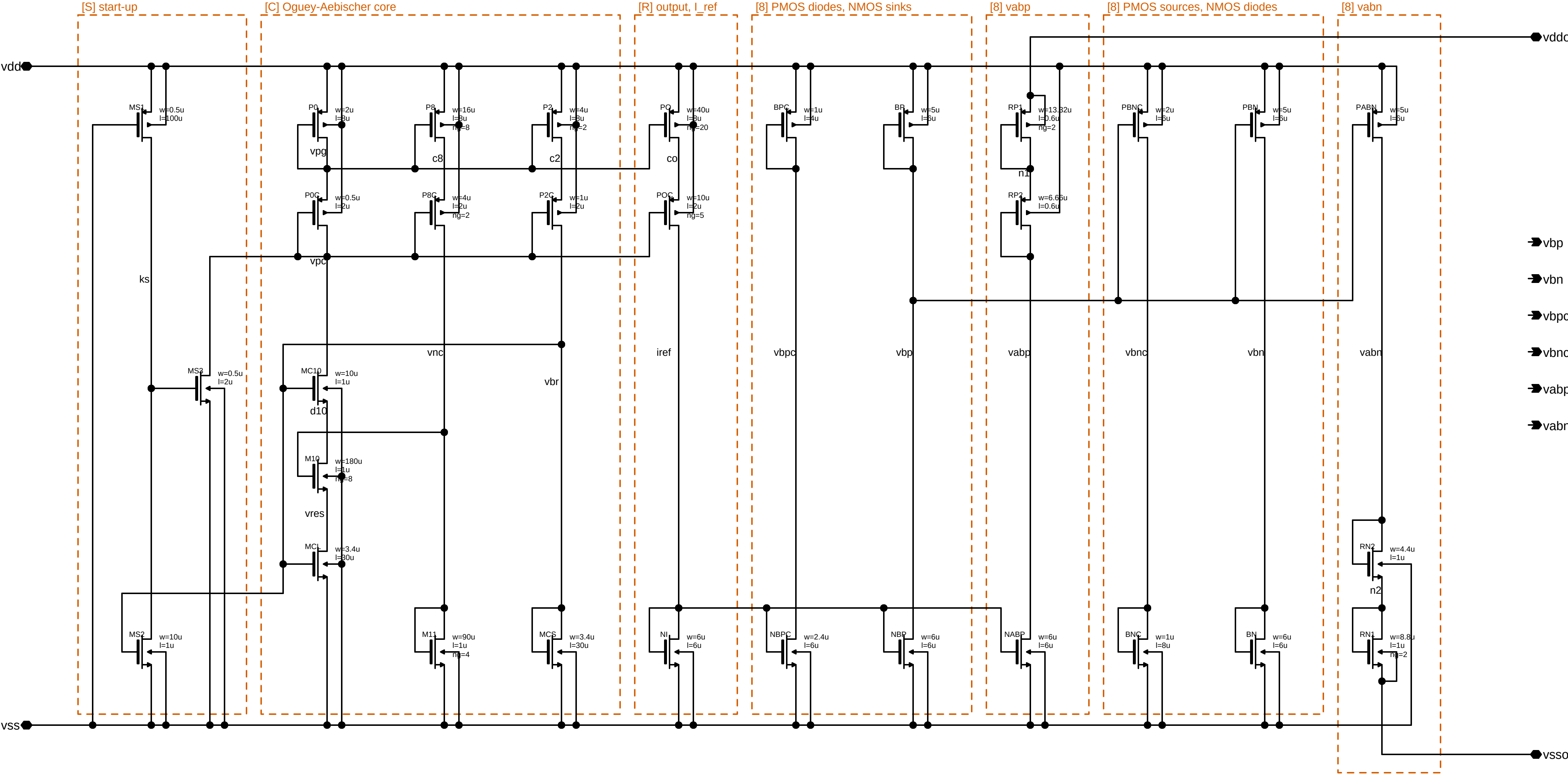


d2s_bias_oa: self-contained bias without resistor or BJT (Oguey-Aebischer core + mirror tree of d2s_bias_in)

core current I0 = 0.25 uA (P0 branch); M11 8 I0, MCS 2 I0; I_ref = 20 I0 = 5 uA into NI



oa_core (Oguey-Aebischer, JSSC 32(7) 1997): no resistor, no BJT.

M11 (8 I0) and M10 (2 x M11 size, I0) share their gate vnc and run in weak inversion, so vres = U_T ln 16 = 72 mV.

MCS (diode, 2 I0) and MCL (triode, I0) share gate vbr and size; MCL drops vres only at one inversion level, so I0 = const * 2 n mu Cox U_T^2 (W/L)_C, independent of V_T. I0 = 0.25 uA; PO / POC deliver 20 I0 = 5 uA into NI.

MC10 cascodes M10; P0 / P0C are stacked diodes, so every mirror output copies P0's V_DS (line sensitivity).

Temperature: I0 ~ mu T^2, about -35 % at -40 C and +33 % at 125 C.

Mirror tree and diodes. The six diodes are those of d2s_bias_ip and set the bias voltages:

vbp: BP 5u/6u, 5 uA (unit tails). vbpc: BPC 1u/4u, 2 uA (mirror cascodes). vabp: RP2 on RP1, 5 uA (ABP, FPL).

vbn: BN 6u/6u, 5 uA (fold sinks). vbnc: BNC 1u/8u, 2 uA (fold cascodes). vabn: RN2 on RN1, 5 uA (ABN, FNL).

PMOS gate lines come from PMOS diodes on vdd, NMOS gate lines from NMOS diodes on vss: bias crosses blocks as currents.

RP1 and RN1 are the class-AB replicas of OP and ON: they sit on the output-stage rails vddo / vssso (separate rails from the right). RP2 keeps its n-well on vdd, like ABP. RN1 needs a local substrate tap ring on vssso, like ON.